Solutions:

1. This question relates to the design of an instrument for measuring bio-potentials from neurons in the brain.

a) Explain why a potential of 70mV can exists across the cell membrane of a neuron when it is at rest.

- Bookwork.
- Explanation of ionic imbalance between the cell membrane of the neuron [1]
- Identification of contributions of sodium and potassium ions [1]
- Saying that this is evaluated using the Nernst potential equation [1]

b) Figure 1.1 shows an AC coupled bio-potential amplifier for measuring neural activity from the brain. The electrodes used have a capacitance CE=60uF and Vsig and Vos denote the measured bio-potential and common mode offset.

i) Give one advantage and one disadvantage of using AC rather than DC coupled bio-potential amplifiers for this application

- Advantage: Remove common mode offset thus allowing large gain in front end. [1]
- Disadvantage: Reduction of input impedance causing potential attenuation of the signal. [1]

ii) C1 and R1 are used to AC couple the input signal to the amplifier. C1 is chosen to be 100pF. Explain why 100pF is a suitable value.

- 100pf is suitable because it is very small relative to the electrode capacitance CE and thus will cause minimum attenuation at the input of the measurement system, that is the capacitor C1. [1]

iii) Derive the input voltage to the amplifier Vin, and show that it is high pass function stating the equation of the cut off frequency stating any assumptions made. To aid you calculations you may use just one branch of the differential input for your analysis.

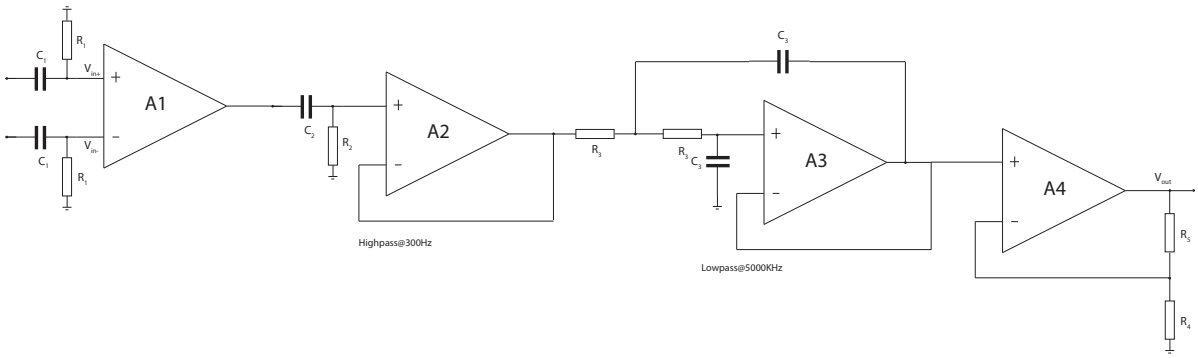
- Assuming C1<<CE we can derive: $V_{in} = (V_{sig} + V_{os}).SR_1C_1/(SR_1C_1 + 1)$ [1]
- $f_c=1/2\pi R_1C_1$ [1]
- Note: You will loose 1 point if you don't state the assumption.

iv) Calculate a suitable value for R1 to give you a high pass cut off at 10 Hz.

- $R_1=1/2\pi C_1 f_c = 1/2\pi(100p)(10)=159M\Omega$ [1]

c) The Bio-potential amplifier in Figure 1.1 is used as part of a monitoring system to detect the action potential signal shown in Figure 1.2. The gain of the Bio-potential amplifier is set to A=1000. The output of the Bio-potential amplifier is band-pass filtered between 0.3KHz and 5KHz with a 20db/decade high-pass followed 40db/decade low-pass filter. A final gain stage is added to maximize the dynamic range between +3V to -3V.

i) Sketch the full schematic of this monitoring system, indicating suitable resistor values for you filters. You may assume all capacitors are, C=100pF for your filters.



- Schematic shown above for complete system (student needs to identify first order high pass and second order low pass (using a sallen key))[4]

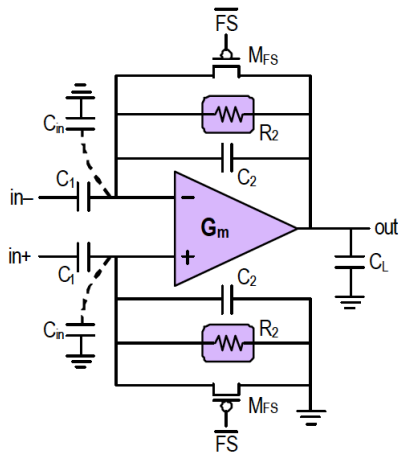
- High pass value $R_2=1/2\pi C_2 f_c = 1/2\pi(100p)(300Hz)=5.3M\Omega$ [0.5]
- Low pass value $R_3=1/2\pi C_3 f_c = 1/2\pi(100p)(5000Hz)=318K\Omega$ [0.5]

ii) Choose suitable gains and resistor values to ensure that the action potential shown in Figure 1.1 is utilizes the maximum dynamic range.

- From figure 1.2 student needs to realize that the 100uV DC is rejected and thus the maximum AC signal passing through is 150uV ptp or 100uV 0top. Realising this the maximum possible gain of the system is 3.3V/100uV=33000. [2]
- Total gain of the system 33000=A1xA2xA3xA4=1000x1x1xA4, thus gain of final stage A4=33[1]
- Thus gain of last stage A4=1+R5/R4=33, therefore R5/R4=32. So we can choose R5=64K and R4=2K. [1]
- Will also accept 150uV ptp as a possible solution giving a gain of 6V/150uV
- Student gets 2/4 points for correct methodology but incorrect input signal.

d) The bio-potential amplifier shown in Figure 1.1 is to be modified to allow a fully integrated implementation in CMOS for recording from a micro-needle array. The requirements are the gain of the modified amplifier is set by a ratio of capacitors and the output and input common mode voltages at DC are the same. Sketch the schematic of the modified bio-potential amplifier.

- Bookwork.
- Sketch of the Harrison amplifier.[2]



2. This question relates to a 2-channel *electrical stimulation circuit for a peripheral nerve interface*.

a) Describe the main benefit and also challenge in the design and implementation of: (i) current-mode; (ii) voltage-mode; (iii) charge-mode electrical stimulation circuits.

[6]

(i) Constant current stimulation allows the charge delivered to be precisely defined through accurate digital timing (since Q=It). [1] However, this comes at the expense of stimulation efficiency, due to a relatively large voltage headroom required to accommodate different electrode impedances. [1]

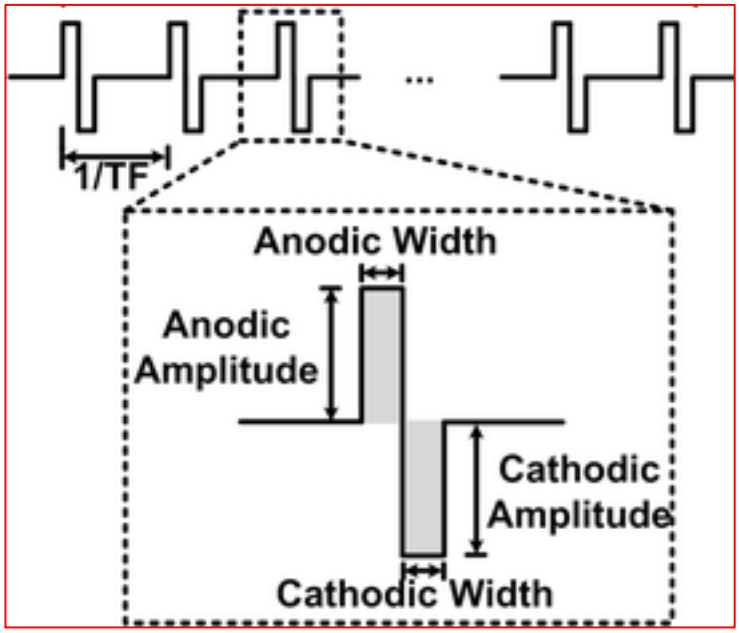
(ii) Constant voltage stimulation boasts good stimulation efficiency as the stimulus is applied directly across the electrodes. [1] However, it is challenging to control the amount of charge delivered to tissue (since the electrode impedance and hence current are unknown, and variable). Thus current monitoring circuits are required. [1]

(iii) Constant charge stimulation boasts good safety as the total amount of charge is limited to what a capacitor can be charged up to. [1] However, there is a challenge of knowing exactly how much charge is delivered to the tissue (i.e. ensuring this charge is completely transferred) and hence efficiency. [1]

b) Sketch a charge-balanced, biphasic stimulation waveform profile to achieve the following parameters:

[4]

Electrode configuration:	Bipolar
Stimulus magnitude:	500nC
Repetition rate:	100Hz per channel
Stimulation strategy:	Continuous interleave sampling



Equal biphasic pulse [1] + Annotating 1/TF=10ms [1]

Anodic width = Cathodic width = 1/(100*4*2) = 2.5ms [1]

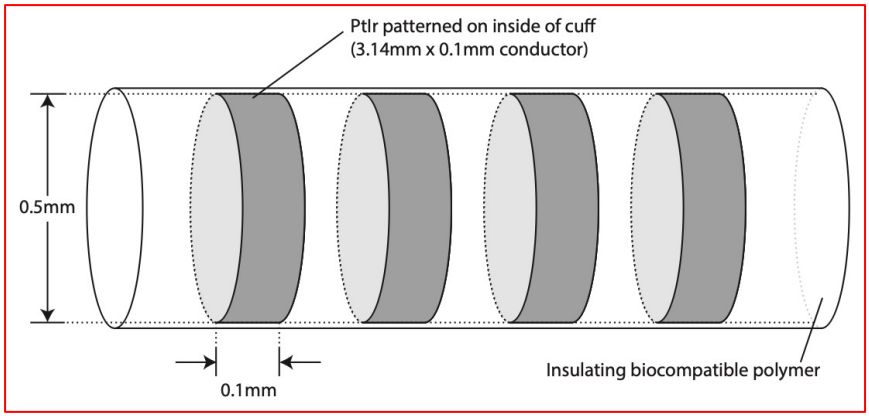
Anodic amplitude = Cathodic amplitude = 100nC/2.5ms = 200uA [1]

c) Determine the conductor geometry for an appropriate cuff electrode to deliver this stimulus to a nerve bundle of diameter 0.5mm considering a platinum iridium conductor (with maximum charge injection capacity of 500uC/cm2).

[4]

Electrode conductor surface area = 500nC/500uC=0.001cm2=0.001*10*10=0.1mm2 [1]

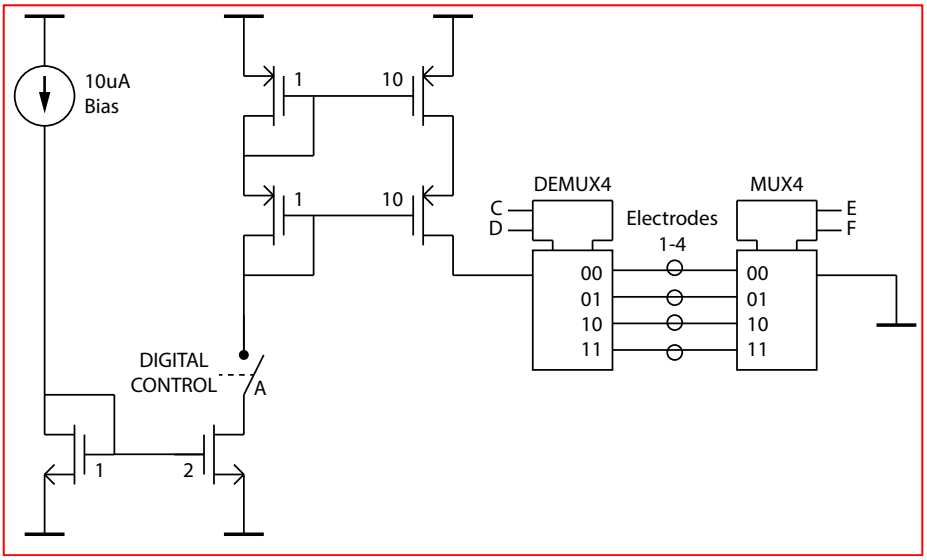
Consider a cylindrical conductor on the inside of a cuff, dimensions = 2 * pi * 0.5mm = 3.14mm x 0.1/3.14= 32um [1] -> 100um (allowing some safety margin, i.e. x3) [1]



[1]

d) Design a circuit for generating the stimulation waveform profile in your answer to (b).

[6]



Scaled current mirrors to save power [1], MUX/DEMUX or H-bridge with current steering switches to reverse current [1], cascode output to increase output impedance of current source since electrode resistance unknown/not given [1].

Truth table/FSM design to generate control signals A, C, D, E, F with 4 states, i.e. S1 and S2 are for channel 1 (electrodes 1-2), S3 and S4 are for channel 2 (electrodes 3-4) [1]

State/Ctrl	A	C	D	E	F
S1	1	0	0	0	1

S2	1	0	1	0	0
S3	1	1	0	1	1
S4	1	1	1	1	0

3. This question relates to the design of chemical sensor instrumentation.

a) Explain the principles of cyclic voltammetry using illustrations where necessary.

[4]

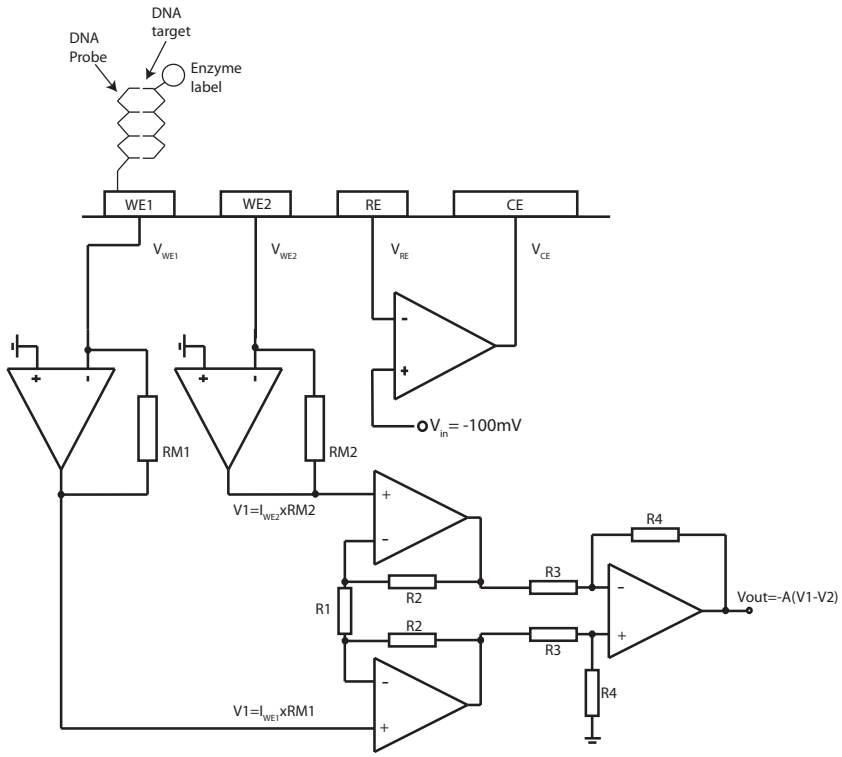
- Bookwork.
- Explanation of Cyclic voltammetry [2]
- Diagram of input and output waveforms [2]

b) Figure 3.1 shows an amperometric sensor used for the detection of DNA hybridization. The sensor works on the principle of a redox reaction confirming hybridization on WE1 at a cell potential of V_{cell}=100mV. The maximum expected output current during hybridization is 10nA. In the same time the sensor drift is expected to be 1uA. An additional electrode, WE2 is added for sensor drift compensation. It is desirable that the output of any instrumentation system use the difference between the hybridization reaction and drift to confirm a reaction.

i. Design instrumentation to interface to this sensor which provides the cell bias, V_{cell} and a suitable readout which can utilize the second electrode, WE2 to increase the signal-to-noise ratio at the output. You are required to show all your schematics and label voltages where necessary and calculate values for all components used stating assumptions where necessary. You may assume any OpAmps used operate between 0-1.8V and you are required to use the full dynamic range of your system.

[8]

- New application of theory. Student needs to identify he can use the standard instrumentation taught and take a differential measurement using a simple difference amplifier.
- Schematic of Potentiostat for correct polarity [1]
- Schematic of transimpedance amplifier with differential readout [4]
- Correct component values [3]
- Transimpedance amplifier – Changes by 1uA , so R=900mV/1uA=900KOhm
- Output Voltage change due to Hybrdisation signal = 900K*10nA=9mV.
- Gain of instrumentation Amplifier= 900mV/9mV=100. So R4=100KOhm, R3=1KOhmn and R1 and R2=10KOhm
- For the schematic
- Loose 1 point for incorrect biasing voltages.
- A potential solution:



ii. Explain why the counter electrode, CE, is larger than the working electrode, WE1.

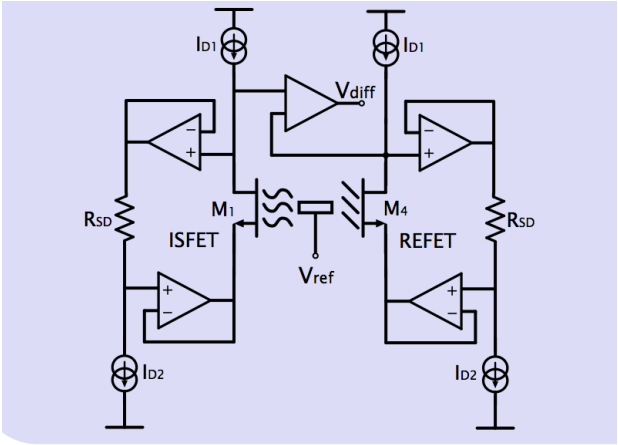
[2]

- Counter electrode has to source double the current to accommodate two working electrodes, therefore it needs to have double the area.

c) Figure 3.2 shows an Ion-Sensitive Field Effect Transistor.

- State two reasons why this sensor better suited for DNA detection than the one shown in Figure 3.1. [2]
  - Scales with Moores law so we can build large arrays reducing cost of sequencing.
  - Allows label free detection without the use of bulky optics.

- The sensor shown in Figure 3.2 is to be used in a DNA detection platform which also includes a Refence-FET (REFET) for background subtraction of any drift. Sketch the schematic of suitable circuit which allows differential measurement from the ISFET and the REFET.
  Differential ISFET Circuit will achieve this:



[4]

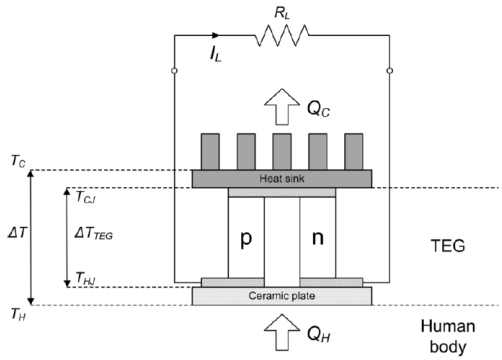
4. This question concerns alternative power supply methods for body worn and implanted medical electronics.

a) Explain, with the aid of a diagram, the structure and principle of operation of a thermo-electric generator (TEG).

[4]

[bookwork]
a) The TEG is a semiconductor structure built from alternating pillars of n-type and p-type material. The basic principle is that as the devices are connected electrically in series but thermally in parallel, carriers diffuse from the hot side to the cold side. This causes electron and hole diffusion to add constructively to the current flow in the device.

[2]



[2]

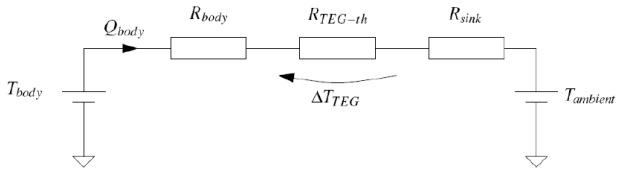
b) A TEG is attached the human body with the aim of powering a wearable device in an ambient temperature that ranges from -2°C to 30°C.

i) Draw and explain an electrical equivalent circuit model to represent the flow of thermal energy through the system from the core of the body to the ambient.

[4]

[bookwork]
b)i) The equivalent circuit model is the same as that types of calculations done for sizing heat sink. Heat flows from the body core temperature, through the body thermal resistance, to the TEG, through the TEG's thermal resistance, and out to the ambient through the thermal resistance to the air (of a heat sink).

[2]



[2]

ii) Assuming the TEG is 100% efficient at converting heat flow into electrical power, calculate the upper limit on the useful electrical output power that can

be obtained for a TEG when the thermal resistance from body-core to TEG is 200 K/W, and the thermal resistance from TEG to ambient is 20 K/W.

[5]

[calculation and interpretation]
ii) The absolute maximum power that can be converted into electrical energy is the heat lost in the TEG thermal resistance (which, if the device is 100% efficient, will be converted into electrical power rather than dissipated as heat). The power converted in the TEG is given by:

$(T_{body}-T_{ambient})/(R_{body}+R_{sink}+R_{TEG})$

[2]

This is maximised when there is a thermal impedance match, i.e. when $R_{TEG}=R_{body}+R_{sink}$

[1]

Hence the power available for conversion into electrical form is (assuming body core temp is 37°C):

$37-T_{ambient}/(2*(220))$= 89mW at -2°C and 16 mW at 30°C.
The useful electrical power is half of these because of the intrinsic electrical impedance of the TEG, i.e. 44 mW and 8 mW.

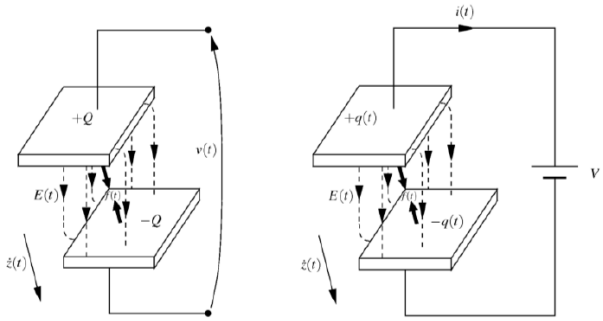
[2]

c) Motion driven energy harvesters are candidates for powering medical electronics. Explain, with the aid of a diagram, two ways in which an electrostatic transducer can be operated to convert mechanical energy to an electrical form.

[4]

c) The electrostatic transducer can either be operated in constant voltage or constant charge. In constant charge mode (shown left), the plates of the transducer are moved relative each other and as capacitance reduces, the voltage increase. Thus, as energy is proportional to the square of voltage, energy increases linearly with position. In constant voltage mode (shown right) the voltage is constant as capacitance reduces with the movement, and hence charge is forced off the transducer into the voltage source, hence there is a net flow of energy into the source.

[2]



[2]

d) Explain the similarities and differences between a regular cored-transformer and an open-cored inductive power transfer system

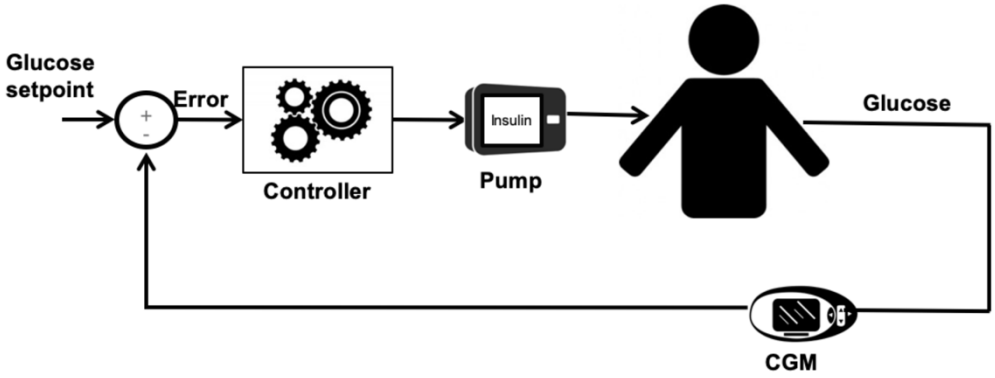
[3

d)
Both systems transfer energy from primary to secondary through magnetic flux linkage and a rate of change of flux in the secondary, caused by the primary, causing voltage across the secondary. However, a regular cored transformer has very low leakage flux and quite high magnetising inductance. An open cored IPT system has very high leakage flux (low coupling factor) and very small magnetising inductances. Hence, the need to have high Q-factors in the case of an IPT system.

5 .This question concerns diabetes management systems.

a) Describe three essential components needed to develop an artificial pancreas (i.e. closed-loop insulin delivery system) and draw the corresponding closed loop block diagram.

Bookwork. Describe a glucose sensor, an insulin pump, a control algorithm



(5 points)

[5]

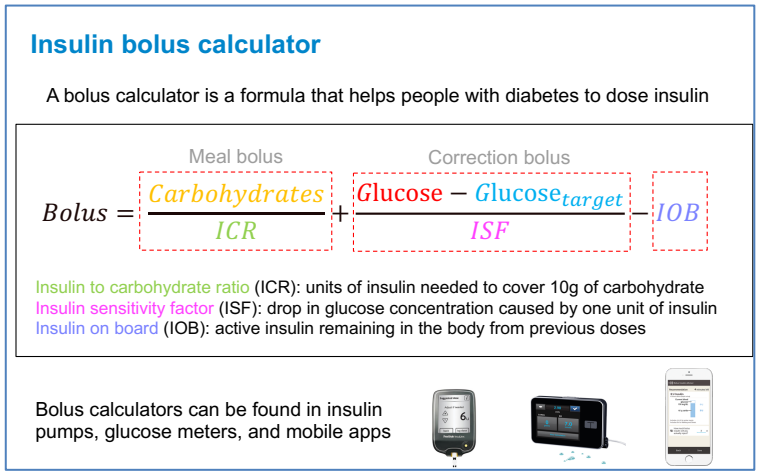
b) State four reasons why an artificial pancreas is still not yet able to achieve optimal blood glucose control (normoglycemia) with the current available technology.

Delay in insulin absorption, sensor inaccuracy, perturbations (meal, exercise), inter- and intra- subject variability
[4 points]

[4]

c) Explain the function of an insulin bolus calculator and state its formula explaining all terms.

Bookwork



[4 points]

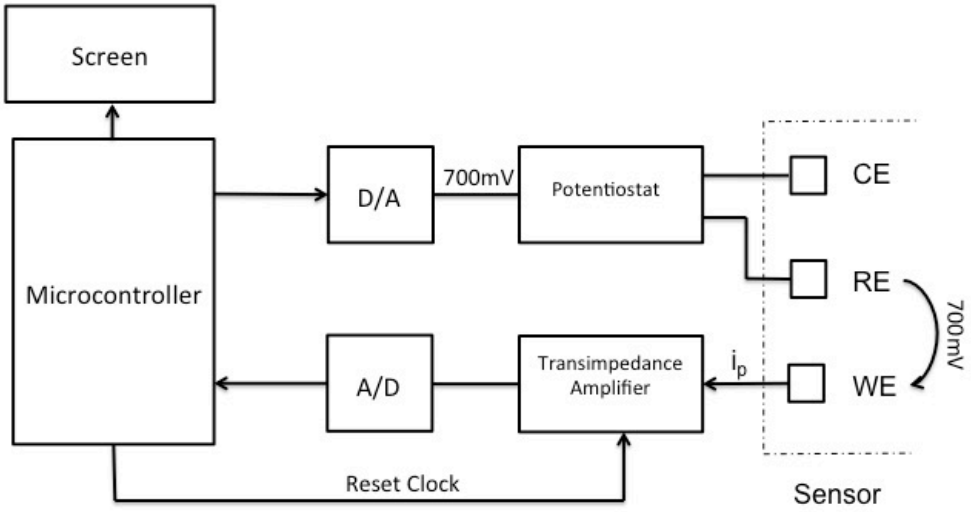
[4]

d) Using the insulin bolus calculator from part c), compute the insulin dose required for the following scenario:
   - Amount of carbohydrate eaten: 70 grams
   - Insulin to carbohydrate ratio: 10 gram/unit
   - Current blood glucose: 10 mmol/L
   - Target blood glucose: 6 mmol/L
   - Insulin sensitivity factor: 2 mmol/L
   - Insulin on board: 1 unit

Solution 70/10 + (10-6)/2 – 1 = 8 units [3 points]

e) Sketch a block level diagram of a glucometer that could be used to measure blood glucose in a person with diabetes.

A possible solution from the lecture notes. Need to specify that Vcell=700mV for glucose sensors and reset signal for trans-impedance amplifier.



[Total 4 points]